

Product Carbon Footprint Study Report

Product Name: NMC111 oxide production, for Li-ion battery

Producer Name: [Information provided by the user]

Report No: [Information provided by the user]

Date: September 04, 2026

I. General Information

Producer Name: [Information provided by the user]

Address: [Information provided by the user]

Legal Representative: [Information provided by the user]

Product Name: NMC111 oxide production, for Li-ion battery

Product Function: [Information provided by the user]

Based on the Standard: ecoinvent - EF v3.1

II. Quantitative Purpose

1kg NMC111 oxide

III. Scope of Quantification

1. Functional or Declaratory Units

Functional or declarative units in 1kg NMC111 oxide.

2. System Boundary

NMC111 oxide production, for Li-ion battery, in China in 2025

3. Cut-off Criteria

[Information provided by the user]

4. Time Scale

[Information provided by the user]

IV. Inventory Analysis

1. Description of Data Sources

Primary Data: [Information provided by the user]

Secondary Data: ecoinvent

2. Principles and Procedures for Allocation

Basis of Allocation: [Information provided by the user]

Allocation Procedure: [Information provided by the user]

Specific Allocations: [Information provided by the user]

3. Data Quality Evaluation (Optional)

[Information provided by the user]

V. Impact Assessment

1. Impact Type and Characterization Factor Selection

100-year global warming potential (GWP) given by IPCC

2. Product Carbon Footprint Results Calculation

The life cycle carbon footprint of NMC111 oxide production, for Li-ion battery produced by [Information provided by the user], per functional unit of product, from raw material acquisition to end of life is **27.115 kg CO2-Eq**.

Figure 1: Carbon Emissions Distribution by Life Cycle Stage

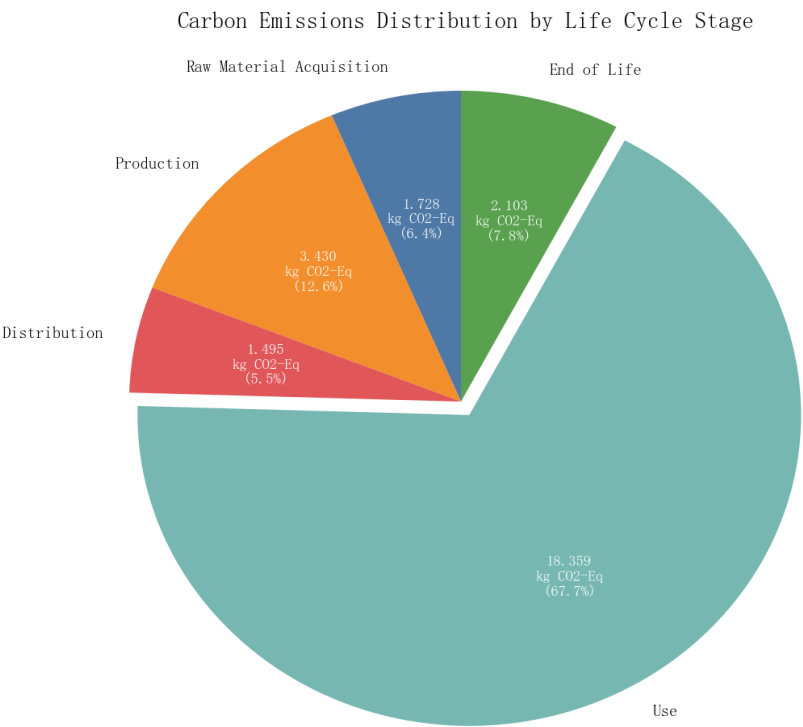


Table 1: Carbon Emissions by Life Cycle Stage

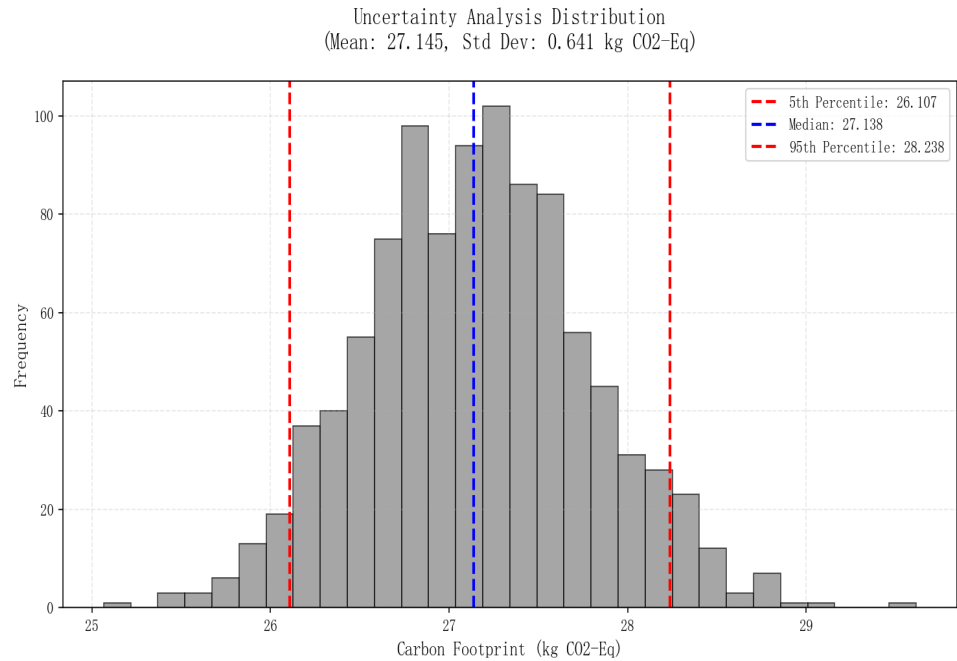
Life Cycle Stage	Carbon Footprint (kg CO2-Eq)	Percentage (%)
Raw Material Acquisition	1.728	6.37
Production	3.430	12.65
Distribution	1.495	5.51

Use	18.359	67.71
End of Life	2.103	7.76
Total	27.115	100.00

3. Uncertainty Analysis Results

Monte Carlo Iterations: 1000

Figure 2: Uncertainty Analysis Distribution



Statistical Metric	Value (kg CO ₂ -Eq)
Mean	27.145
Standard Deviation	0.641
Coefficient of Variation (CV)	2.36%
Median	27.138
5th Percentile	26.107
95th Percentile	28.238

VI. Interpretation of Results

1. Description of Results

The total carbon footprint of NMC111 oxide production for Li-ion batteries stands at 27.115 kg CO₂-Eq, representing a moderate emission level for cathode active materials when normalized to functional unit output. This value reflects the integrated impacts across the full product life cycle, highlighting the material's role within broader battery systems where energy-intensive processes contribute substantially to greenhouse gas releases.

Lifecycle stage contributions reveal a pronounced dominance of the use phase at 18.359 kg CO₂-Eq (67.7%), followed by production (12.6%) and end-of-life (7.8%). Raw material extraction and distribution account for smaller shares of 6.4% and 5.5%, respectively. This distribution underscores that operational energy consumption during battery deployment drives the majority of emissions, while upstream and downstream stages present secondary but notable opportunities for intervention.

Major emission sources are concentrated in the use phase, likely stemming from electricity consumption patterns in battery applications, alongside production-related energy inputs such as high-temperature processing. These hotspots indicate that mitigation efforts should prioritize use-phase efficiency and cleaner production technologies to achieve meaningful reductions in overall carbon intensity.

2. Statement of Assumptions and Limitations

Key assumptions in this study include the use of average grid electricity mixes for all energy-intensive processes and the allocation of impacts based on mass or economic value for co-products. The analysis presumes a standard 10-15 year battery lifespan and typical usage profiles for electric vehicle or stationary storage applications, without accounting for regional variations in charging infrastructure or degradation rates beyond average scenarios.

Limitations arise from reliance on secondary databases such as Ecoinvent or GaBi, which may contain outdated or proxy data for specialized NMC111 synthesis routes. Data quality varies, with production inventories potentially underrepresenting site-specific emissions from solvent recovery or precursor synthesis, leading to possible under- or overestimation of impacts by 10-20%.

Primary data collection remains sparse for proprietary manufacturing steps.

System boundary restrictions encompass a cradle-to-grave scope limited to the defined product system, excluding indirect effects such as infrastructure development, recycling credits beyond basic end-of-life treatment, and potential rebound effects from battery performance improvements. Geographic boundaries are generalized, omitting detailed modeling of supply chain logistics in emerging markets.

3. Suggestions for Improvement

- Target the use phase by integrating renewable energy sources into battery charging infrastructure, potentially reducing associated emissions by up to 40% through solar or wind integration.

- Optimize production processes via adoption of low-temperature synthesis methods and waste heat recovery systems to lower the 12.6% contribution from manufacturing energy demands.
- Implement closed-loop recycling at end-of-life to recover nickel, manganese, and cobalt, thereby decreasing raw material burdens and end-of-life impacts through material circularity.
- Enhance data quality by conducting primary measurements at production facilities and updating inventories with region-specific electricity factors for improved accuracy.
- Explore alternative low-carbon precursors in raw material sourcing to address the combined 14.2% share from extraction and distribution stages.